WORKED SOLUTION

STEP II 2013, Question 6

Pure • Sequences

STEP II 2013, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q6

This question has a full guided treatment in *Beyond STEP*, Section 2.2, *The Golden Ratio as a Limit*: https://beyondstep.keithmcnulty.org/ch02/golden-limit.html

The key idea

The terms $1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \ldots$ jump alternately above and below their eventual limit, so the whole sequence is not monotonic and the theorem cannot be applied to it directly. Taking every second term, however, gives two monotonic sequences. Part (i) is the tool that shows this: it says that the step $u_{n+2} - u_n$ has the same sign as the previous step $u_n - u_{n-2}$, because the denominator is positive.

Solution

First note that every u_n is positive: $u_1 = 1 > 0$, and if $u_n > 0$ then $u_{n+1} = 1 + 1/u_n > 1 > 0$. So we never divide by zero, and every factor such as $1 + u_n$ is positive.

Part (i)

Using $(*)$ twice,

$$u_{n+2} - u_n = \frac{1}{u_{n+1}} - \frac{1}{u_{n-1}} = \frac{u_{n-1} - u_{n+1}}{u_{n+1}u_{n-1}}, \qquad u_{n+1} - u_{n-1} = \frac{1}{u_n} - \frac{1}{u_{n-2}} = \frac{u_{n-2} - u_n}{u_n u_{n-2}}.$$

(The second needs u_{n-2} to exist, which is why $n \geq 3$.) Substituting the second into the first, the two minus signs cancel:

$$u_{n+2} - u_n = \frac{u_n - u_{n-2}}{(u_n u_{n+1})(u_{n-2}u_{n-1})}.$$

Finally, multiplying $(*)$ by u_n gives $u_n u_{n+1} = u_n + 1$, and likewise $u_{n-2}u_{n-1} = u_{n-2} + 1$. Hence

$$u_{n+2} - u_n = \frac{u_n - u_{n-2}}{(1 + u_n)(1 + u_{n-2})},$$

as required.

Part (ii)

We use induction. The claim $1 \le u_1 \le 2$ is true since $u_1 = 1$. Suppose $1 \le u_n \le 2$. Taking reciprocals of positive numbers reverses the inequalities, so $\frac{1}{2} \le 1/u_n \le 1$, and therefore

$$\tfrac{3}{2} \le u_{n+1} = 1 + \frac{1}{u_n} \le 2,$$

which certainly gives $1 \le u_{n+1} \le 2$. By induction, $1 \le u_n \le 2$ for all n.

Part (iii)

The odd terms. We have $u_1 = 1$ and $u_3 = \frac{3}{2}$, so $u_3 - u_1 > 0$. For odd $n \ge 3$, part (i) says $u_{n+2} - u_n$ is $u_n - u_{n-2}$ multiplied by the positive number $1/\big((1+u_n)(1+u_{n-2})\big)$, so it has the same sign. By induction every step $u_{n+2} - u_n$ (n odd) is positive: the sequence $u_1, u_3, u_5, \ldots$ is increasing. By (ii) it is bounded above by 2, so by the theorem it tends to a limit L, with $1 \le L \le 2$ since all its terms lie in $[1, 2]$.

The even terms. Here $u_2 = 2$ and $u_4 = \frac{5}{3}$, so $u_4 - u_2 < 0$, and the same argument shows that $u_2, u_4, u_6, \ldots$ is decreasing. It is bounded below by 1.

> **Ask yourself** The theorem is about increasing sequences that are bounded above. How can we use it for a decreasing sequence?

Apply it to the negatives: $-u_2, -u_4, -u_6, \ldots$ is increasing and bounded above by -1, so it tends to a limit, and hence so does $u_2, u_4, \ldots$; call its limit M. Again $1 \le M \le 2$.

Finding the limits. Two steps of $(*)$ give a recurrence linking terms of the same parity:

$$u_{n+2} = 1 + \cfrac{1}{1 + \cfrac{1}{u_n}} = 1 + \frac{u_n}{u_n + 1} = \frac{2u_n + 1}{u_n + 1}.$$

> **Ask yourself** Why may we simply replace u_{n+2} and u_n by L?

Along the odd n, both u_n and u_{n+2} tend to L. The right-hand side is a continuous function of u_n (its denominator is at least 2), so it tends to $(2L+1)/(L+1)$. Limits are unique, so

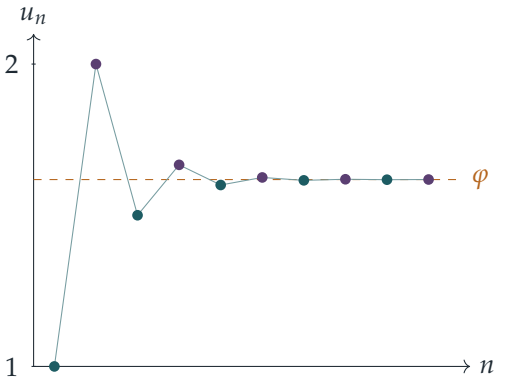
$$L = \frac{2L+1}{L+1} \quad \Longrightarrow \quad L^2 + L = 2L + 1 \quad \Longrightarrow \quad L^2 - L - 1 = 0,$$

so $L = \frac{1}{2}(1 \pm \sqrt{5})$. As $L \ge 1$, we must have $L = \frac{1}{2}(1 + \sqrt{5})$. Exactly the same equation holds for M, with $M \ge 1$, so

$$\lim_{k \to \infty} u_{2k-1} = \lim_{k \to \infty} u_{2k} = \tfrac{1}{2}\big(1 + \sqrt{5}\big).$$

The two limits might have been different (in which case the whole sequence would oscillate), but the equation has only one root that is at least 1.

The whole sequence. Write $\varphi = \frac{1}{2}(1+\sqrt{5})$ and let $\varepsilon > 0$. There is a K_1 such that $|u_n - \varphi| < \varepsilon$ for all odd $n > K_1$, and a K_2 such that the same holds for all even $n > K_2$. Then $|u_n - \varphi| < \varepsilon$ for every $n > \max(K_1, K_2)$, so $u_n \to \varphi$.



The odd terms (teal) climb towards φ and the even terms (plum) descend towards it.

Starting from $u_1 = 3$

The conclusion does not change. Now $u_1 = 3$, $u_2 = \frac{4}{3}$, $u_3 = \frac{7}{4}$, $u_4 = \frac{11}{7}$. The argument needs only small adjustments.

- Part (i) used only $(*)$ and the positivity of the terms, which still holds, so the identity is still true.
- Part (ii) fails at $n = 1$, but $1 \le u_2 = \frac{4}{3} \le 2$, and the inductive step is unchanged, so $1 \le u_n \le 2$ for all $n \ge 2$. Also $u_1 = 3 \ge 1$, so every term is at least 1.
- This time $u_3 - u_1 = -\frac{5}{4} < 0$ and $u_4 - u_2 = \frac{5}{21} > 0$. So the odd terms are decreasing and bounded below by 1, and the even terms are increasing and bounded above by 2: the roles are swapped, but both still converge.
- The recurrence $u_{n+2} = (2u_n + 1)/(u_n + 1)$ still holds and both limits are at least 1, so both are again φ.

So the sequence still tends to $\frac{1}{2}(1+\sqrt{5})$.

Checks

With $u_1 = 1$ the terms are $1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \ldots$, ratios of consecutive Fibonacci numbers, and $u_{10} = \frac{89}{55} \approx 1.61818$, already very close to $\varphi \approx 1.61803$. With $u_1 = 3$ the terms are $3, \frac{4}{3}, \frac{7}{4}, \frac{11}{7}, \frac{18}{11}, \ldots$ (ratios of Lucas numbers), and these also approach 1.61803. You can also check the identity in (i) with $n = 3$: $u_5 - u_3 = \frac{1}{10}$, and $\dfrac{u_3 - u_1}{(1+u_3)(1+u_1)} = \dfrac{1/2}{5} = \frac{1}{10}$.

Common slips

- Assuming the whole sequence converges and solving $L = 1 + 1/L$ straight away. That gives the right number, but only after convergence has been proved, which is the point of the question.
- Forgetting the root $\frac{1}{2}(1 - \sqrt{5})$: it must be ruled out, here by $L \geq 1$.
- Applying the theorem to the decreasing even subsequence without saying how (for example, via $-u_{2k}$).

Beyond STEP

This is a model for a standard first-year analysis argument: the monotone convergence theorem, subsequences, and the fact that if the odd and even subsequences converge to the same limit then so does the whole sequence. In fact u_n is the nth convergent of the continued fraction $1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}$, and the map $x \mapsto 1 + 1/x$ is a contraction near φ, which is why any positive start leads to φ. Abbott's *Understanding Analysis* treats monotone sequences and subsequences carefully, and Davenport's *The Higher Arithmetic* covers continued fractions.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.